

Recommended optimized Santum AI architecture:

Tier	Input Cap	Output Cap	Monthly Tokens	RAG Level	Best Use
Free	80 words	80 words	4,000	Light	short support
Standard	100 words	100 words	100,000	Light	CBT support
Premium	120 words	120 words	200,000	Moderate	deeper support

Core Model Setup

Function	Recommended Model
Main counselling response	GPT-4.1
Memory summarisation	GPT-4.1 Mini
Session summaries	GPT-4.1 Mini
Tags / routing / analytics	GPT-4.1 Nano
Safety classification	GPT-4.1 Mini or dedicated moderation layer

User-facing therapy:

GPT-4.1 because:

*emotional quality matters,
tone matters,
pacing matters,
safety consistency matters.*

Background tasks:

GPT-4.1 Mini for:

*summarization,
memory compression,
analytics,
RAG filtering,
moderation assistance.*

Ultra-light infrastructure tasks:

GPT-4.1 Nano for:

*labels,
routing,
metadata,
tiny summaries,
classifications.*

Memory Architecture

Use:

<i>Memory Type</i>	<i>What It Stores</i>
<i>Recent context</i>	<i>last 3–6 exchanges</i>
<i>Session summary</i>	<i>current emotional theme</i>
<i>Long-term memory</i>	<i>recurring issues, goals, preferences</i>
<i>Safety memory</i>	<i>risk flags, escalation notes</i>

Avoid sending full chat history.

RAG Architecture

Free Tier — Light RAG

Use only when needed.

Retrieve:

safety guidance
basic CBT snippets
Santum AI disclaimers
escalation instructions

Recommended:

1 chunk
50–100 words
only on relevant turns
Standard Tier — Light RAG+

Retrieve:

light CBT tools
grounding exercises
anxiety/depression support snippets
Santum platform information
referral/escalation rules

Recommended:

1–2 chunks
80–150 words each
not every message

Premium Tier — Moderate RAG

Retrieve:

structured CBT modules
psychoeducation snippets
journaling prompts
coping exercises
user-specific therapeutic goals
Santum referral pathways

Recommended:

2–3 chunks
100–200 words each
only when clinically useful

Estimated Costs

<i>Tier</i>	<i>Retail</i>	<i>Likely Monthly Cost/User</i>	<i>Margin</i>
<i>Free</i>	<i>\$0</i>	<i>~\$0.03–0.12</i>	<i>acquisition cost</i>
<i>Standard</i>	<i>\$5</i>	<i>~\$0.80–2.50</i>	<i>good</i>
<i>Premium</i>	<i>\$7</i>	<i>~\$2–5.50</i>	<i>viable but tighter</i>

Estimated Usage

<i>Tier</i>	<i>Sessions</i>	<i>Talk Time</i>
<i>Free</i>	<i>~1 short session</i>	<i>10–30 mins</i>
<i>Standard</i>	<i>~8–20 sessions</i>	<i>4–9 hrs</i>
<i>Premium</i>	<i>~18–45 sessions</i>	<i>8–18 hrs</i>

Best Overall Design

Use GPT-4.1 as the therapeutic voice, but keep RAG controlled.

Best rule:

Base model handles emotional conversation. RAG only supports safety, structure, policy, and specific therapeutic exercises.

This gives Santum AI the best balance of:

low cost,
good CBT quality,
fast responses,
user safety,
scalable memory,
and commercially viable pricing.

Achieving Better Cost Efficiency By:

*aggressive summarized memory,
short recent context windows,
and lightweight structured memory,
short inputs,
short outputs,
and conversational pacing
light RAG for safety/policy
moderate RAG for CBT exercises
avoid heavy RAG except for rare admin/legal/clinical protocol cases.
RAG retrieval is selective,
RAG snippets are short, and the AI synthesizes them conversationally.*